Research statement

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My work centers on moduli spaces of sheaves on complex surfaces of Kodaira dimension zero. My broader interests include algebraic and arithmetic geometry and their connections with representation theory and mathematical physics. The main directions of my research are the following.

1. Topology and geometry of moduli spaces of semistable sheaves on bielliptic surfaces. Moduli spaces of sheaves often reflect the geometry of the underlying surface. For suitable primitive Mukai vectors and generic polarizations, K3 surfaces give irreducible holomorphic symplectic moduli spaces [Muk84, O'G97, Huy97]; on abelian surfaces, Albanese fibers play a similar role [Yos01]. The remaining minimal surfaces of Kodaira dimension zero, namely Enriques and bielliptic surfaces, behave differently because their canonical bundles are nontrivial torsion line bundles. Following Saccà's work on Enriques surfaces [Sac19], my preprint [Pis26] studies moduli spaces of sheaves on bielliptic surfaces via relative compactified Jacobians of linear systems.

For a bielliptic surface S, I start with a primitive rank-zero Mukai vector with nonzero Euler characteristic and sufficiently positive first Chern class, and also choose a generic polarization. Write M for the distinguished component of $M_{H,S}(\mathbf{v})$, $M(L)$ for its distinguished component with determinant L, and F for their common Albanese fiber. My first main result determines their first two Betti numbers:

	b_1	b_2
F	0	3
$M(L)$	2	4
M	4	9.

The support morphism $h : M(L) \to |L|$ for rank zero sheaves also provides a surjection $\pi_1(S) \twoheadrightarrow \pi_1(M(L))$. Fourier–Mukai transforms and Bridgeland wall crossing extend the Betti-number calculations to selected components for *admissible* primitive positive-rank vectors (i.e. those that can be reduced to this rank-zero range), again for generic polarizations.

I also prove that $H^2(F, \mathbb{C})$ is pure of weight two and Hodge type $(1, 1)$; in particular, F has no reflexive holomorphic two-forms. When the rank-zero model admits a further Fourier–Mukai reduction to rank one, F has a finite quasi-étale cover (étale in codimension one) that is a Calabi–Yau variety. The reflexive tangent sheaf of F apperas to be strongly stable. If, in addition, the polarization remains generic on the canonical abelian cover, the selected moduli component is smooth and noncanonically birational to a product of $\mathrm{Pic}^0(S)$ and a Hilbert scheme of points on S.

The proofs combine the geometry of the support morphism in rank zero with monodromy and mixed Hodge structure theory. The results describe both the abelian contribution to the moduli spaces and the geometry of their Albanese fibers. In ongoing work with Howard Nuer, we study irreducibility of the moduli spaces and fixed determinant fibers, and the Beauville–Bogomolov type of Albanese fibers beyond the rank-one reduction. Further questions concern vectors not covered by these results, the kernel of the fundamental-group map, and the exact orders of the canonical bundles.

2. Weak Brill–Noether theory on Enriques and bielliptic surfaces. The weak Brill–Noether problem asks whether a general stable sheaf in $M_H(\mathbf{v})$ has at most one nonzero cohomology group. It is a surface analogue of classical Brill–Noether theory. The answer depends on the geometry of the surface and can be studied using Bridgeland wall crossing. General results are known for K3 and abelian surfaces [CNY23, CNY25], while the Enriques and bielliptic cases remain open in general.

In ongoing joint work with Howard Nuer, Edoardo Mason, Roberto Vacca, Huishi Yu, and Filippo Papallo, we prove the property for large families of Mukai vectors on Enriques surfaces, with numerical conditions expressed in terms of the Cossec–Dolgachev function. The proof combines Fourier–Mukai transforms, Bridgeland wall crossing, and a numerical analysis of totally semistable walls. We next aim to treat the complementary range, construct counterexamples, and compute the cohomology of a general stable sheaf. I also plan to study analogous questions on bielliptic surfaces.

The following sections expand on the research outlined on the first page.

1 Topology and geometry of moduli spaces of semistable sheaves on bielliptic surfaces

Setting and notation

Recall that a bielliptic surface is a smooth projective minimal complex surface S with $\kappa(S) = 0$, $q(S) = 1$, and $p_g(S) = 0$. Its canonical bundle is nontrivial torsion, its canonical cover $\pi : X \to S$ is an abelian surface, and it carries two elliptic fibrations, one of which is its Albanese morphism. These features provide both the geometry of the support morphism and the Fourier–Mukai transforms used below.

For a Mukai vector $\mathbf{v} = (r, c, s)$, write $\mathbf{v}^2 = c^2 - 2rs$ and let $M_{H,S}(\mathbf{v})$ denote the moduli space of Gieseker H-semistable sheaves with Mukai vector $\mathbf{v}$. The vector retains the integral first Chern class, including its torsion part; numerical conditions use its image in $\mathrm{Num}(S)$. In the standard basis A_0, B_0, with $A_0^2 = B_0^2 = 0$ and $A_0 \cdot B_0 = 1$, set $\mathrm{nt}(D) = \min\{a, b\}$ for $D \equiv aA_0 + bB_0$. A vector is *admissible* if it has rank zero with $\mathrm{nt}(c_1(\mathbf{v})) \geq 3$, or has positive rank and a Fourier–Mukai autoequivalence sends it to such a vector.

In rank zero, $M^\circ_{H,S}(\mathbf{v})$ denotes the distinguished irreducible component containing sheaves locally free on their integral support, and $M_{H,S}(\mathbf{v}, L)$ is the corresponding fixed-determinant component. For positive rank, fix a Fourier–Mukai reduction to rank zero, the wall-crossing data, and the corresponding component $M^\circ_{H,S}(\mathbf{v})$. $M_{H,S}(\mathbf{v}, L)$ then denotes any irreducible component of its determinant fiber over L. All statements use these fixed choices. Let F be an Albanese fiber of $M_{H,S}(\mathbf{v}, L)$. Under the hypotheses below, it is also an Albanese fiber of $M^\circ_{H,S}(\mathbf{v})$.

The integer λ_S denotes the minimal multisection degree of the Albanese fibration of S, and $\ell(\mathbf{v})$ is the divisibility of $\pi^*\mathbf{v}$ in the algebraic Mukai lattice of the canonical cover. For primitive rank-zero vectors with positive square, $\lambda_S = \ell(\mathbf{v}) = 1$ characterizes vectors whose Fourier–Mukai orbit contains a rank-one vector. Throughout, b_i denotes the dimension of singular cohomology with rational coefficients.

Main results

Theorem 1.1 ([Pis26, Theorem A])**.** *Let (S, H) be a polarized bielliptic surface and let $\mathbf{v}$ be a primitive admissible Mukai vector. If $\mathbf{v} = (0, c_1(L), \chi)$ has rank zero, assume $\chi \neq 0$. Assume that H is $\mathbf{v}$-generic. Then:*

	b_1	b_2
F	0	3
$M_{H,S}(\mathbf{v}, L)$	2	4
$M^\circ_{H,S}(\mathbf{v})$	4	9.

If $\mathbf{v} = (0, c_1(L), \chi)$ has rank zero, there is a surjective homomorphism

$$\pi_1(S) \twoheadrightarrow \pi_1\big(M_{H,S}(\mathbf{v}, L)\big).$$

Moreover, if $\lambda_S = \ell(\mathbf{v}) = 1$ and π^*H is $\pi^*\mathbf{v}$-generic, then $M^\circ_{H,S}(\mathbf{v})$ is smooth, irreducible, and projective of dimension $\mathbf{v}^2 + 1$ and is noncanonically birational to

$$\mathrm{Pic}^0(S) \times \mathrm{Hilb}^{\mathbf{v}^2/2}(S).$$

Theorem 1.2 ([Pis26, Theorem B])**.** *Under the hypotheses of the first assertion of Theorem 1.1, the Hodge structure on $H^2(F, \mathbb{C})$ is pure of weight two and of type $(1, 1)$:*

$$H^2(F, \mathbb{C}) = H^{1,1}(F).$$

In particular, $H^0(F, \Omega^{[2]}_F) = 0$, where $\Omega^{[2]}_F = (\Omega^2_F)^{**}$. If, moreover, $\lambda_S = \ell(\mathbf{v}) = 1$, there is a finite quasi-étale cover $\widetilde{F} \to F$ such that $\widetilde{F}$ is a Calabi–Yau variety in the sense of [GGK19, Definition 1.3]. Consequently, the reflexive tangent sheaf $\mathcal{T}_F$ is strongly stable.

The Albanese morphism separates the abelian contribution to a moduli space from the geometry of its fibers. The two theorems describe both parts: the first computes their low-degree topological invariants, while the second rules out holomorphic two-forms on the fibers and, in the rank-one-orbit case, identifies a Calabi–Yau cover. The surjectivity result for the fundamental group gives information beyond the first Betti number, but leaves the kernel undetermined.

Geometric approach

The rank-zero case is the geometric core. For $\mathbf{v} = (0, c_1(L), \chi)$ under the assumptions of Theorem 1.1, the support morphism

$$h : M_{H,S}(\mathbf{v}, L) \longrightarrow |L|$$

has the compactified Jacobian of degree $\chi - \chi(\mathcal{O}_C)$ as its fiber over an integral curve C. Over the locus of smooth curves, monodromy determines the invariant cohomology classes. Mixed Hodge structures then control the effect of adding the discriminant and the remaining boundary, which allows us to compute the cohomology of these moduli spaces. The fundamental-group argument adapts the method of Leibman and Saccà to the case of a bielliptic surface whose fundamental group is infinite and nonabelian.

For positive-rank sheaves, Fourier–Mukai reduction maps the Mukai vector and stability condition to rank zero. Subsequent Bridgeland wall crossing relates the selected component to a rank-zero Gieseker moduli space. By Nuer's results [Nue25], these moduli spaces have terminal local complete intersection singularities and a torsion canonical bundle. Minimal-model theory provides isomorphisms in codimension one. Since these spaces are local complete intersections and their singular loci have codimension at least four, we can compare their low-degree cohomology. The Calabi–Yau cover is obtained by comparison with Albanese fibers of Hilbert schemes and the construction of Oguiso and Schröer [OS11].

Further directions

The results suggest the following questions, discussed in [Pis26] and in my thesis.

1. **Irreducibility and determinant fibers.** In ongoing work with Howard Nuer, we study when $M^{\circ}_{H,S}(\mathbf{v}) = M_{H,S}(\mathbf{v})$ and when $M_{H,S}(\mathbf{v}, L)$ is the entire determinant fiber. In the rank-zero case with $\lambda_S = \ell(\mathbf{v}) = 1$, irreducibility of the full moduli space is already known under the hypotheses above. The question is to understand the remaining cases and the determinant fibers.
2. **Geometry of Albanese fibers and canonical bundles.** Does the conclusion of Theorem 1.2 about a Calabi–Yau cover hold for all primitive admissible vectors? If not, which abelian, Calabi–Yau, or irreducible holomorphic symplectic factors occur after a finite quasi-étale cover in the Beauville–Bogomolov decomposition? Understanding this is part of the same joint project with H. Nuer. A related question is the exact order of the canonical bundle: can it be expressed in terms of the surface type and the Mukai vector, and how does it change upon fixing the determinant or passing to an Albanese fiber?
3. **Extending the range and determining fundamental groups.** The rank-zero cases with $\mathrm{nt}(c_1(\mathbf{v})) < 3$ or $\chi = 0$ remain outside the present results. For positive-rank vectors not reducible to the admissible rank-zero range, standard reductions often lead to sheaves of rank two or three. These lower-rank moduli spaces may be studied directly. Another question in the rank-zero range already covered in [Pis26] is what is the kernel of $\pi_1(S) \twoheadrightarrow \pi_1(M_{H,S}(\mathbf{v}, L))$? Is the resulting fundamental group independent of L and the generic polarization?

AI disclosure: The problem formulation, main ideas of the proofs and mathematical exposition in [Pis26] were human-generated and developed by the author and his advisor during PhD studies H. Nuer. AI tools (such as ChatGPT 5.6 Sol and 6 Astra by OpenAI) were used to assist with LaTeX formatting, locating sources for cited results, checking English grammar, and checking some mathematical results and computations concerning the orbits of Mukai vectors.

2 Weak Brill–Noether theory on Enriques and bielliptic surfaces

Setting and main result

Fix an unnodal polarized Enriques surface (X, H) and a Mukai vector $\mathbf{v} = (r, c, a/2)$ with $c \cdot H > 0$ and $\chi(\mathbf{v}) = (r+a)/2 \geq 0$. The weak Brill–Noether problem asks whether a general stable sheaf $E \in M_{H,X}(\mathbf{v})$ has at most one nonzero cohomology group. In this range, $H^2(X, E) = 0$, so the problem reduces to proving $H^1(X, E) = 0$.

The numerical bound in our main result is expressed through the Cossec–Dolgachev function

$$\phi(L) = \inf\{|L \cdot E| : 0 \neq E \in \mathrm{Num}(X),\ E^2 = 0\},$$

which measures intersections with isotropic divisor classes.

Theorem 2.1 ([NMP+26])**.** *Let* $\mathbf{v} = (r, dH, a/2)$ *be a Mukai vector on a polarized Enriques surface* (X, H) *such that* $r \geq 0$*,* $a \leq 0$*, and*

$$d \geq \frac{|a + r|}{2\phi(H)}.$$

Then $M_{H,X}(\mathbf{v})$ *satisfies the weak Brill–Noether property.*

Idea of the proof

We use Fourier–Mukai transforms to study higher-cohomology vanishing and generic global generation for E and $E(K_X)$. These conditions hold precisely when the derived dual of the transformed object is a coherent sheaf. We study this object by varying the Bridgeland stability condition toward the Gieseker chamber. Potential obstructions arise from totally semistable walls, whose destabilizing factors satisfy numerical restrictions on their Mukai vectors. This reduces the problem to a finite lattice-theoretic analysis. The Cossec–Dolgachev function bounds intersections with isotropic classes and excludes the potential obstructions in the range of the theorem, yielding the required cohomology vanishing.

Further directions

1. **The complementary range.** The case $a > 0$ remains open: the lattice-theoretic argument for $a \leq 0$ does not directly apply, so we may need other methods.
2. **Cohomology and counterexamples.** When potential obstruction walls occur, we aim to compute the cohomology of a general stable sheaf using the stable factors at these walls and the resulting extensions. These computations would help us construct counterexamples to weak Brill–Noether in this range. This requires distinguishing genuine cohomological obstructions from walls that do not affect the general sheaf.
3. **Bielliptic surfaces.** I also plan to study analogous questions on bielliptic surfaces using relative Fourier–Mukai transforms along the elliptic fibrations. The nontrivial Albanese morphism affects cohomology and generic global generation, so the Enriques argument cannot be transferred directly.

AI disclosure: In this joint project, AI tools (such as Gemini by Google and ChatGPT by OpenAI) were used for numerical simulations and searching for integer solutions to the inequalities describing destabilizing walls.

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